21 Electrical Installations

By Victor Maggioli Sr., Graham Elvis, & Lawrence (Larry) M. Thompson

Topic Highlights

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Motto for Electrical Installation: *"It depends upon the application."*

21.1 Introduction

This chapter is about electrical installations (EI) in industrial facilities. It is designed to provide the necessary information to ensure automation processes do not fail due to faulty electrical installation practices.

Covered in this chapter are the basics of grounding, both for safety and signal noise reduction, including salient aspects such as power quality and uninterrupted power supply (UPS) systems, along with a cursory look at electrical circuit protection and enclosures.

21.2 Scope

This section addresses key electrical installation criteria necessary to achieve a reliable electrical installation in an industrial manufacturing facility utilizing automated processes. These EI criteria include power quality, plant-level electrical distribution, grounding, electromagnetic coupling (EMC), power line conditioning, and noise characteristics.

Please note that this section does not consider EI requirements for office buildings or large computer facilities, such as those found in system information departments.

21.3 Grounding and Bonding

Grounding plays an important role in the ability of EI to function safely and properly.

Properly grounded electrical installations allow protective devices—for example, fuses, circuit breakers, metering, ground fault interrupters (GFIs), lightning arrestors, and surge protectors to function properly. A properly grounded and designed EI will place the circuit in a safe, open (no power) condition during:

- overloads,
- short circuits (before equipment explodes),
- ground faults, and
- surges (caused by, for example, a lightning strike).

Personnel safety is outlined in the National Fire Protection Association (NFPA) National Electrical Code (NEC) and the Canadian Electrical Code (CEC) and should be completely understood and integrated into the EI design.

21.3.1 Grounding

Electrical systems that are grounded shall be connected to the earth in a manner that will limit the voltage imposed by lightning, line surges, or unintentional contact with higher voltage lines and that will stabilize the voltage to earth during normal operation. (NEC Article 250-4(A)(1))

21.3.2 Grounded

Grounded means connected to the earth or to some conducting body that serves in place of the earth. (NEC Article 100)

21.3.3 Grounding Conductor

The conductor used to connect non-current-carrying metal parts of equipment, raceways, and other enclosures to the system grounded conductor, the grounding electrode conductor, or both, at the service equipment or at the source of a separately derived system. (NEC Article 100)

21.3.4 Bonding

Bonding is the permanent joining of metallic parts to form an electrically conductive path which will assure electrical continuity and the capacity to conduct safely any current likely to be imposed. (NEC Article 100)

21.4 Grounding Systems

NEC requires the system ground be established (preferably at the service entrance), and is concerned primarily with personnel safety and fire protection. Establishing a signal zero volt reference must work in conjunction with the NEC. The zero volt reference is necessary to reduce noise and ground loops, which are not the goals NEC was designed to specify. Although the two purposes are generally in harmony, at times there are issues in satisfying both objectives. While we will cover some grounding considerations and practices in this chapter (this is not the definitive grounding reference) it should be understood that the chapter is primarily concerned with how the grounding system augments the performance of automation systems.

The main thrust of all signal grounding systems is to develop a workable ground plane. Due to the high frequency signals that abound in modern control systems, the aim is to establish an equipotential point—effectively a large area of the same impedance—with multiple grounds including the service entrance point. There are many ways to do this, which will be addressed at the appropriate point in the chapter.

21.4.1 Grounding Electrode

The primary grounding element for a small facilities ground is a single electrode consisting of a rod, pipe, or plate (made electrodes). Note that if a single ground electrode (at the service entrance or entry point of a separately derived ground) has a resistance greater than 25 ohms it should be augmented by an additional electrode. However, the NEC does not say what to do if the resistance is still above 25 ohms. In fact, most EI should have a ground resistance of 10 ohms (MIL HDBK 419A) and in practice, 5 ohms or less.

21.4.2 Grounding Triads

A single ground rod is scarcely able to establish an equipotential plane or even an effective ground. In fact, in most industrial systems a triad is used. A triad consists of three ground rods (a minimum of 3 meters [10 ft] long—longer in high resistive soils) driven in the ground in a triangular configuration a minimum of 3 meters (10 ft) apart. They are then bonded together and used as a single rod.

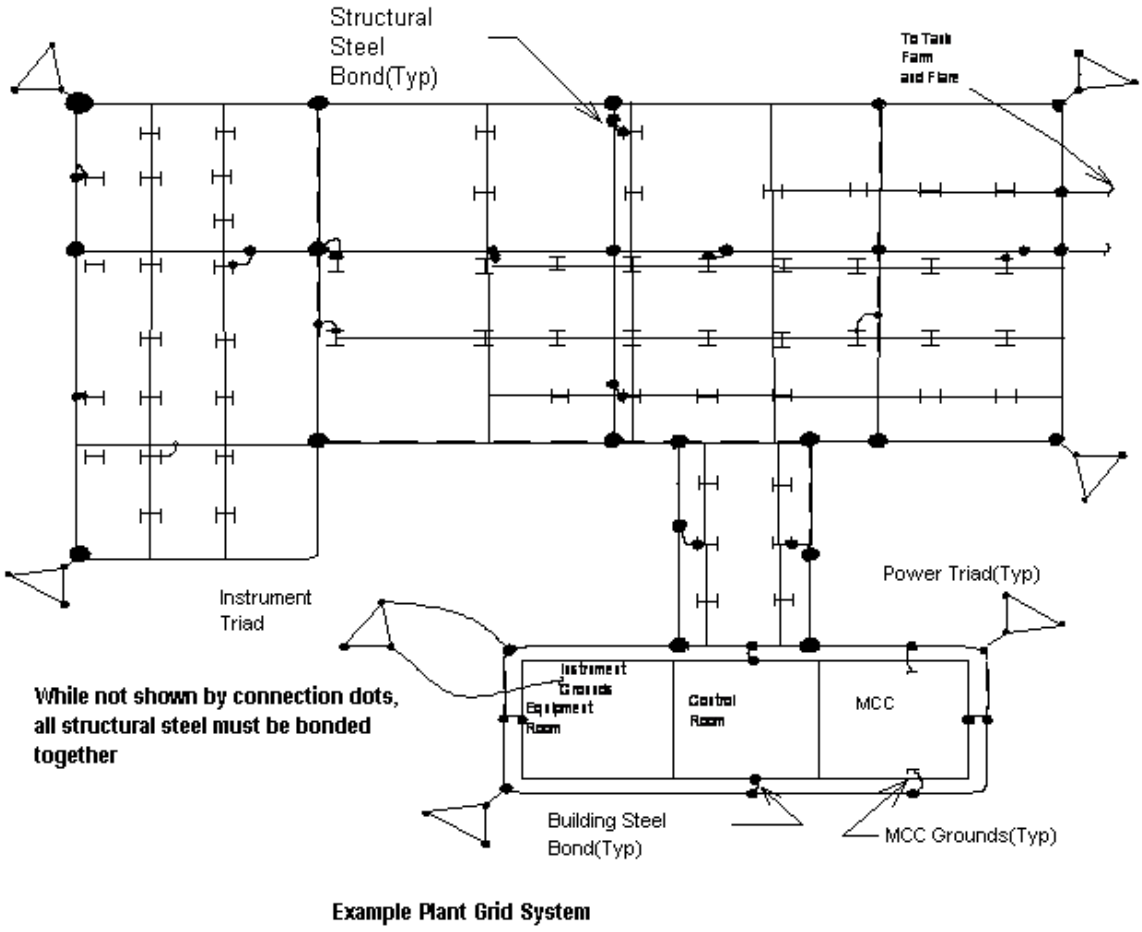


Figure 21-1: Triads and Building Grounds

21.4.3 The Equipotential Plane

The equipotential plane is a surface underneath (typically) the operating area consisting of conducting thin sheets or copper screen bonded together at multiple points typically by triads. It includes the service ground. This area has very low impedance and is effectively one large ground area.

The conducting media used for an equipotential plane can be (a) a copper grid embedded in the concrete floor, (b) a raised metal floor such as computer floor, (c) a subfloor of aluminum, copper, phosphor bronze screen, or sheet metal laid underneath the floor tile or carpet, or (d) a ceiling grid above

tile equipment. The grid openings should not be larger than 1/20 wavelength at the highest frequency of concern, up to four inches. As a design objective the grid openings should not be larger than four inches." (Mil HDBK 419A)

At higher frequencies, this large conducting surface under the equipment to be grounded will present much lower characteristic impedance than a single wire. This is due to characteristic impedance (Z0) being a function of L/C. As the amount of capacitance to earth increases, Z0 will decrease.

Typically, the capacitance to ground of the conducting plane to earth is higher than that of a single conductor. If the size of the conducting plane is increased and allowed to encompass more area, the capacitance will increase. Also, the inductance generated by length of the conducting sheet will decrease with width further decreasing Z0.

Significantly increasing the dimensions of a conducting sheet (as it will be with the conducting plane) causes the characteristic impedance to approach a very low value. It will remain quite low throughout a large portion of the spectrum. This will establish an equipotential reference plane for all equipment bonded to it. Any "noisy" conductor through or along the equipotential ground plane will have its fields constrained between the conductor and the ground plane. (The material on the equipotential plane has been extracted and paraphrased from Mil HDBK 419A.)

21.4.4 Separately Derived Grounds

In a standard service entrance point solid ground, the neutral and safety ground are tied directly to the grounding electrode at this point, which is the only point they connect within the system. This is done so if there is a ground fault in the system, the circuit interrupter will open (circuit breaker or fuse), protecting the circuit. In some processes this could pose a problem, particularly if the ground fault causes an interruption to equipment controlling a process. Halting a process immediately is not generally a good idea. Typically a separately derived ground is used.

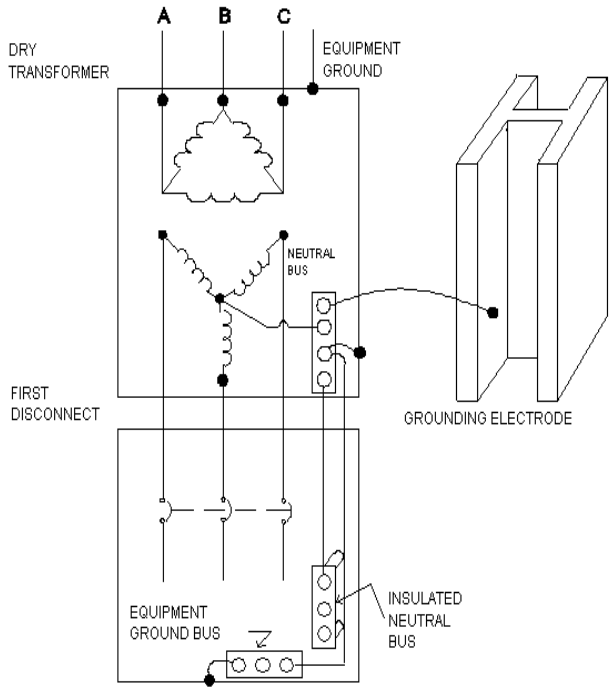


Figure 21-2: Separately Derived System

Note that the input from the delta transformer has no metallic connection to the grounded system (a generator could be used in place of a transformer), so the ground is separately derived. In place of the solid wire to the grounding conductor, an impedance or resistance could be used that would then allow a ground fault to exist without interrupting the system. The only requirement is a ground fault detection system (and a competent workforce) must be in place.

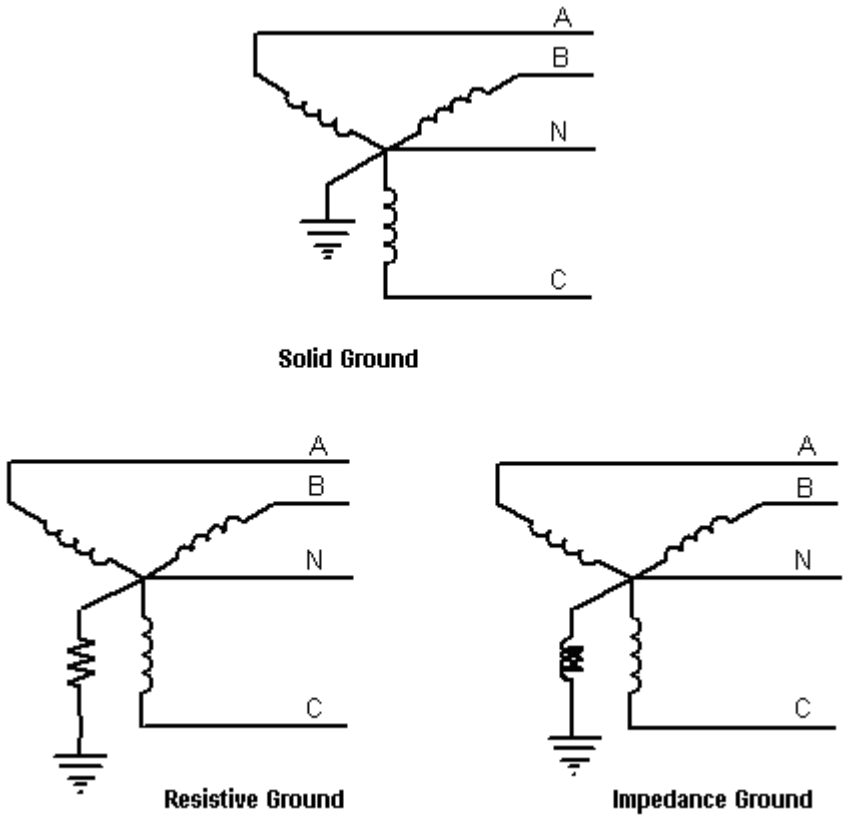


Figure 21-3: Ground Types

21.4.5 Other Recommendations

Steel framed buildings shall be designed so the steel is electrically continuous.

Single point grounds are recommended for instrument power panel(s) distribution transformer(s) servicing programmable electronic (PE) devices.

All communication wiring between ground planes, such as buildings and systems with different grounding approaches, shall be nonmetallic—for example, fiber optic with nonmetallic strength members.

All PE devices and related distribution equipment shall have surge protection sized to:

- protect the device or equipment immediately on the load side of the surge protector
- coordinate with the upstream device, such as substation lightning arrestors

21.4.6 Equipment

The EI shall use equipment suitable for the application (e.g., NE Code, Article 500). However, PE equipment requires additional considerations to achieve the desired safety and functionality. These considerations include:

- mounting to eliminate vibration and ground currents—e.g., PE device mounting on an enclosure backplane via nonconductive vibration isolation mounts
- assembly exactly per manufacturer's instructions to eliminate electromagnetic compatibility (EMC) problems
- wiring routing to allow proper heat dissipation in the cabinet housing PE devices
- single point grounds where possible
- isolation from EMC sources
- voltage level separation to eliminate EMC problems (see manufacturers installation instructions)

21.4.7 Enclosure Grounding

Cabinet grounding is illustrated in the diagram below. There is no need for an insulated (from the cabinet) grounding strip in the bottom of the cabinet, because the cabinet is bonded to the equipotential plane and is part of that plane. Signal returns and shields are connected as shown in the diagram.

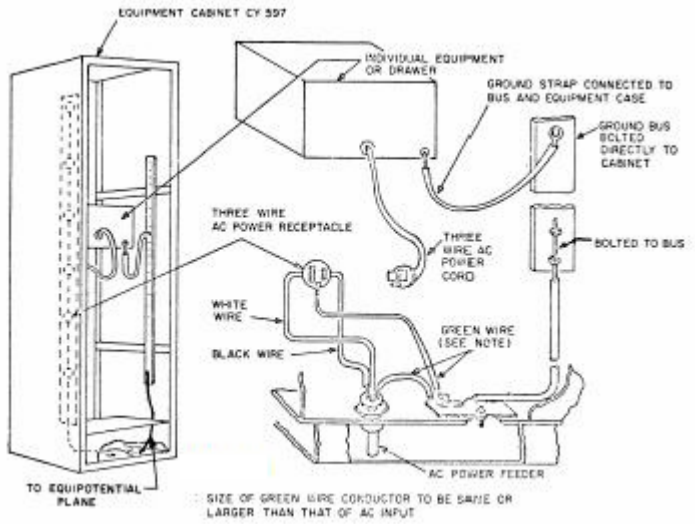


Figure 21-4: Typical Cabinet Grounding (Graphic extracted from MIL HDBK 419A)

21.5 Ground Loops

Ground loops are caused by a difference in potential between ground points in a circuit that causes undesired current to flow between the ground points, hence the term "ground loop." A ground loop does not have to involve current flowing in earth ground.

This is one reason why the shield should only be grounded at one end. There are other problems if the signal return itself is also grounded at both ends. The transmitter could very well be a thermocouple transmitter with the couple mechanically (and hence electrically) bottomed in the thermal well to make good thermal conductivity. There are many solutions, although the simplest is just to ensure the shield is only grounded in this case at one end. By grounding the shield at one end only we are using the "single point ground," which is quite effective for circuits with noise below 1 MHz.

In most cases grounding the shield at one end only effectively reduces the noise coupling acting as a Faraday shield between the signal line and ground. However, what about high frequency (HF)—above

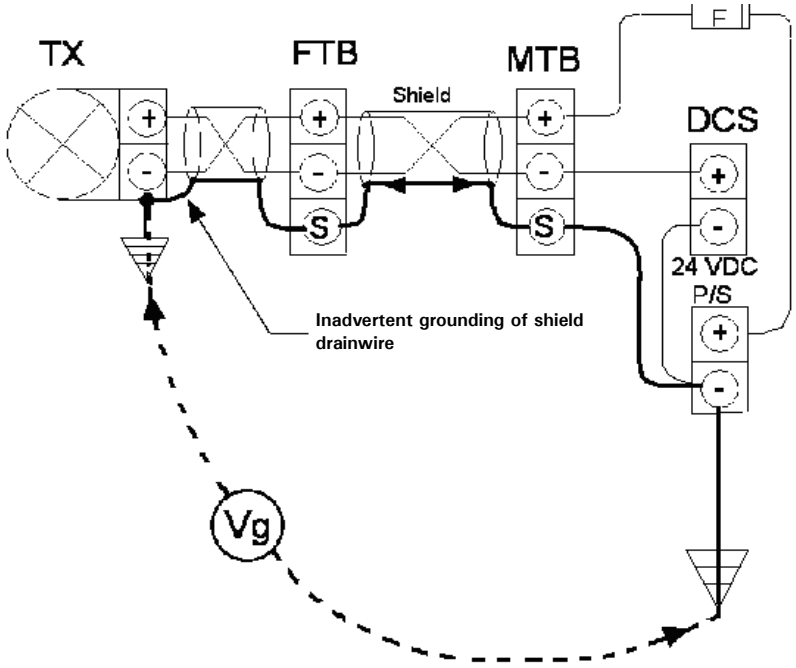


Figure 21-5: Typical Ground Loop

1 MHz—noise? The purpose of the equipotential plane is to have a 0 V reference over a large area, and in the case of HF a multiple-point ground is desired.

The use of shielded cable is a well-known technique for combating noise; however, when HF noise is a problem, it may be better to ground the shield at both ends.

A shielded cable, with both ends of the shield bonded to the relevant equipment chassis, acts as a 1:1 transformer at HF, the shield being the "primary" and the conductors being the "secondary." Any voltage applied between the two ends of the shield (the difference between the two pieces of equipment) will generate an identical voltage in the conductors. The polarity is such that the induced voltage exactly cancels the applied or common mode noise voltage. This effect is so strong that a 9-m (29-ft) length of feedback cable will typically reduce the common mode noise by a factor of 60dB (1000:1), and a 90-m (290-ft) length by 80dB (10000:1). When only one end of the shield is grounded this attenuation effect will be completely lost.

21.6 Noise Reduction

Although an entire treatise on shielding and other noise reduction techniques is beyond the scope of this chapter, there are certain installation details that if followed will go a long way toward reducing electrical noise.

21.6.1 The Noise Model

Noise is a communication, similar in characteristics to our desired communications. Noise has a source, a path, and a receptor. We have discussed grounding at some length, and having a good ground (0 volt reference) is essential to noise reduction.

Basically, if you can stop the noise at the source you are way ahead. Most of our efforts are placed on interrupting the path or protecting (shielding) the receptor (because it is very difficult to determine the source, when the source is offensive, and if we have the jurisdictional authority to move or diminish the source—lightning comes to mind).

21.6.2 Interrupting the Path

Physical separation is (usually) one of the least expensive methods for reducing noise. Radiated noise is a square law adherent, meaning the field strength of radiated noise falls off as the square of the distance from the source. If you are twice as far from the source, the radiated power reaching you is one-quarter of what it would be if you were only half that distance. Table 21-1 is a list of noise generators, their intensity of noise and what cabling to use. Please note that simply running the different level signals in different conduits will break up the paths by allowing less coupling of noise to your signal leads.

Table 21-1: Noise Sources

	Zone Category			
Cable	**Very Noisy**	**Noisy**	**Quiet**	**Cable Shielded**
Three phase between drive and filter (shielded)		X		X
Three phase between drive and filter (unshielded)	X			
PWM drive/inverter – motor power (shielded)		X		X
PWM drive/inverter – motor power (unshielded)	X			
External dump resistor (unshielded)	X			
External dump resistor module (metal-clad) – bus connections		X		
External dump resistor module (metal-clad) – fan connections		X		
Motor feedback			X	
Three-phase supply power		X		
Single-phase supply power		X		
Hydraulic/pneumatic solenoids (suppressed)		X		
AC motor contactors	X			
Solid-state relay output (zero crossing)		X		
PLC digital I/O		X		
Dedicated drive inputs (except registration)		X		
Limit switches		X		
Pushbuttons		X		X
Proximity switches (except registration)		X		X
Photo electric cell		X		
24 VDC relay contact (suppressed)		X		
Transformer indicator lamp		X		
Encoder buffer board			X	X
Data/communications (See note 2)			X	X
Encoder			X	X
High-speed registration			X	X
PLC analog I/O			X	X
PLC high-speed counter input			X	X

The system builder's objective must be to minimize the coupling mechanism between noise sources and their receptors. There are five techniques commonly employed to deal with electrical noise in systems:

- High-frequency bonding
- Shielding
- Segregation
- Filtering
- Suppression (of mechanical contacts)

21.6.2.1 High-frequency Bonding

Lengths of wire, however heavy, will exhibit significant impedance at HF, making them unsuitable for HF bonding. For example, a 500-mm (19.7-in) length of wire behaves as an efficient antenna at the Amateur Radio frequency of 144 MHz. It forms a quarter-wave antenna (a quarter of the 2-m wavelength) that matches the 50-ohm output impedance of the transmitter to that of free space, 377 ohms. Thus, a simple length of wire can appear as anything between 50 and 377 ohms, depending only on length and frequency.

The purpose of HF bonding is to ensure all the metalwork of a system is at the same voltage at HF; then, if two pieces of equipment are linked by a cable, the cable will not be carrying any "common mode" voltage (the voltage between the two ends of the cable).

The solution is to use the "Equipotential Ground Plane" principle: every point on a large, flat, conductive surface will be at the same potential at all frequencies, that is, at almost zero impedance between all points. This is a well-known practice in printed circuit board design. One layer is left largely intact and used as a reference for all the devices on the board.

We can easily achieve this within a control cabinet by using the panel itself, which is much easier if the panel is zinc-plated steel. If paint is used it must be scraped off under each mounting point and is then liable to corrode, thus negating the bonding effect.

Multiple panels, if adjacent, may be bonded top, middle, and bottom with wide, short straps to extend the ground plane over the whole suite.

PWM drive power cables should *always* be shielded if possible. At high powers this may become impractical, so conduit may be used to create a shield. However, it must be continuous, ferrous, and terminated at both the panel and motor frame.

21.6.2.2 Separation

Clearly if the source and receptor are widely separated (the path is lengthened) they will not interact. A 200-mm (7.9-in) distance is enough to drastically reduce interaction, but even that amount is impractical in a typical, crowded panel. The solution is to define areas of the control panel as "quiet" and "noisy" zones. Each is reached via "quiet" and "noisy" wire ways that can be most easily identified by the use of different colors, say, black for noisy and gray for quiet. (See Table 21-1.)

21.6.3 Protecting the Receptor

If all else fails, we will fortify the receptor itself. We can actually shield the target (receptor). There are two forms of shielding: ferrous and nonferrous. *Ferrous* means the material is magnetic. Iron and steel are ferrous; aluminum is not. Surrounding equipment in nonferrous materials (such as with a cabinet made of aluminum) protects it from electromagnetic radiation—radiated signals. Reducing the electrostatic field of an electromagnetic radiation reduces the signal itself. However, in industry many fields are created by large currents, and these tend to be magnetic fields (H). Nonferrous shielding has very

little effect on magnet fields. Ferrous materials, on the other hand, provide a lower reluctance path for magnetic fields and hence protect from them. Because most ferrous materials are also conductors, they protect against the electrostatic (E) field as well.

When protecting a piece of equipment or even a facility, all conductors in and out of the facility must have filters and other techniques applied to prevent noise from being conducted into (or in some cases, out of) the protected area.

21.7 Electrostatic Noise

Electrostatic noise is generated when the built up charges on insulating surfaces discharge. Lightning is a dramatic example. Wind currents in the updraft of a thundercloud charge the bottom of the cloud negatively (and hence the top of the cloud, where the ionized atoms end up, positively). The earth provides an oppositely charged surface. When the potential difference is high enough a leader stroke ionizes the path, and the main stroke then attempts to equalize the charges in the familiar lightning stroke, followed by a thunderclap due to the heated air around the stroke or strokes.

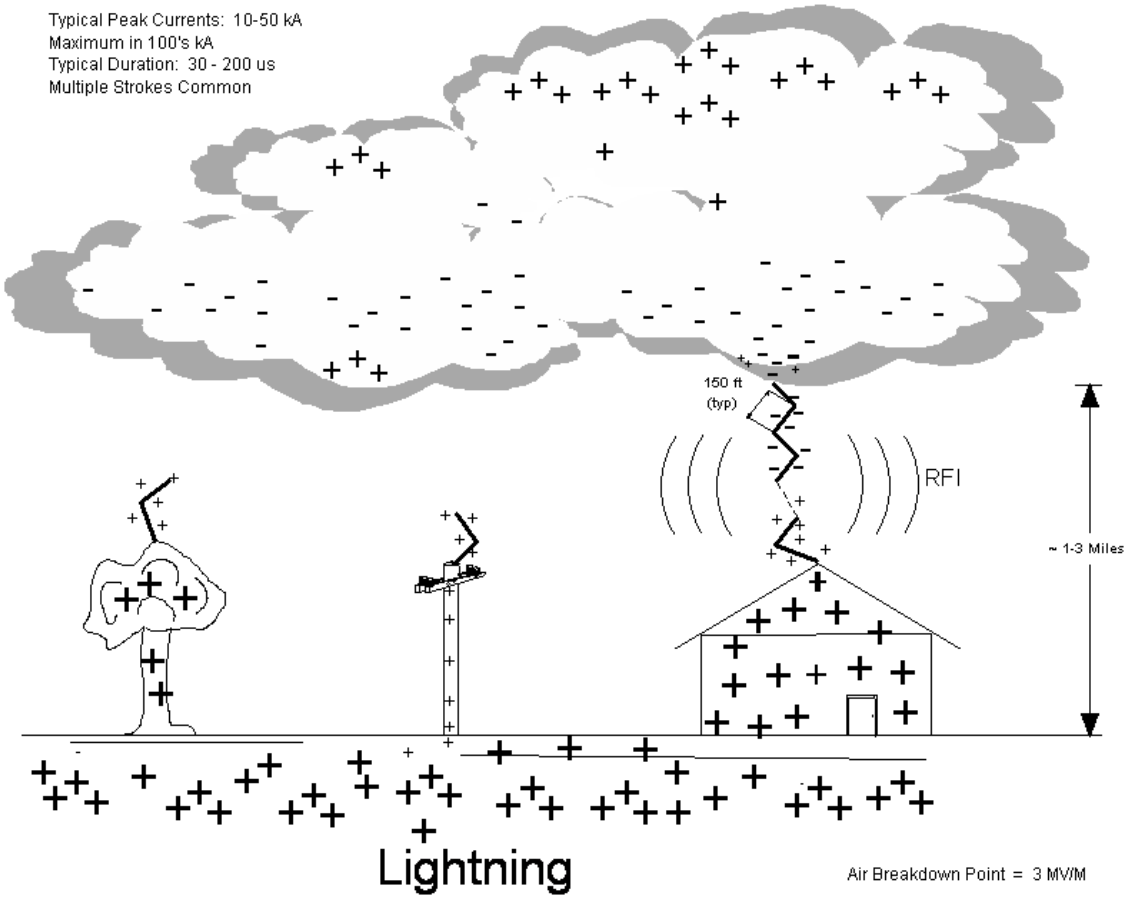


Figure 21-6: Lightning as an ESD

Similarly, on a much smaller scale, whenever there is a charge difference and it becomes high enough for the distance involved, an electrostatic discharge (ESD) will occur, destroying unprotected electronic circuits. The main concern in this chapter, though, is with lightning and its radiated and conducted effects.

21.7.1 Radiation from an ESD

When an ESD occurs, it generates extremely wideband electromagnetic energy with quite high (momentary) energy, resulting in a wideband pulse. This noise can enter control systems as a pulse, or spike, causing unanticipated results. Noise shielding and good grounding will reduce the susceptibility of circuitry to this noise.

21.7.2 Conducted Noise from an ESD

The major concern with lightning (other than a direct hit) is the noise spike introduced and conducted through the system. Long runs of conductors parallel to the earth's surface (unless encased in ferrous conduit) are subject to having ESD noise coupled due to induction. A cloud-to-cloud strike will produce a mirror earth current paralleling the cloud-to-cloud strike. If this is parallel to the conductor's, a current will be induced in the conductors and will have then entered the system. Other impulse noise is disruptive, but seldom destroys equipment.

21.7.3 Protection from ESD

There are a number of methods to protect systems from the effects of lightning: proper grounding, shielding, and active protection in the form of surge suppressors. Other concerns, such as noise spikes and random pulses produced by equipment are reduced by noise suppression techniques such as shielding and active noise reduction circuitry. Power quality concerns (of which ESD protection is one such) include a number of methods to ensure that the power quality will enable correct equipment operation and longevity. Surge suppressors are used for just that reason and are one of the primary protections from lightning as an ESD.

21.8 Surge Suppressors

Surge suppressors go by a number of names, dependent upon their construction and materials. Two of the best known names are:

- Transient Voltage Surge Suppressor (TVSS)
- Surge Protection Device (SPD)

21.8.1 TVSS

The major active components of transient voltage surge suppressors are:

- Gas tube, which is also called a gas discharge or gas surge arrestor
- Solid state
  - SAS are silicon avalanche diodes and go by commercial names such as Transorbs.
  - Silicon controlled rectifiers (SCRs) are used as crowbars in some surge suppressors to handle higher currents than SAS can.
- Metallic oxide semiconductors (MOVs) are also known as voltage-dependent resistors or varistors. MOVs are subject to deterioration from repeated surges and should have a fuse in series with the MOV and preferably some form of indication or alarm upon MOV failure.

Table 21-2: General Capacities of TVSS

Type	Capacity	Response Time
Gas Tube	High	> 5nS
Solid State	Low - Medium	< 5nS
MOV	Medium	< 5nS

A commercial device called a RAV manufactured by Comm-Omni International is a combination of a gas tube and a varistor.

21.8.2 TVSS Circuitry

The active components alone do not make a complete TVSS but need additional components to work correctly. Most TVSS are hybrid devices (a combination of active components), and some are made up of only passive components. The following is an example of TVSS circuitry.

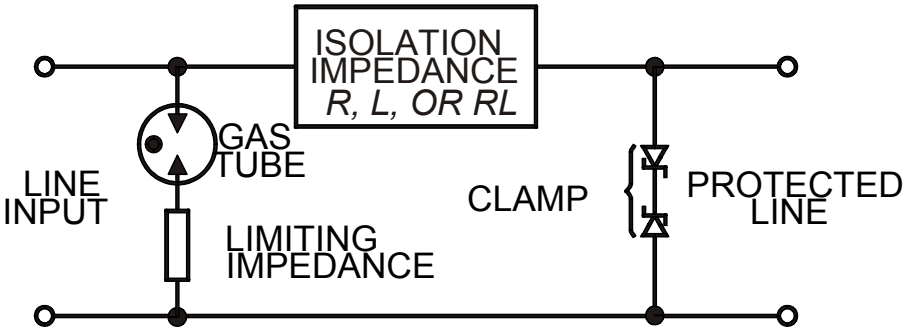


Figure 21-7: A Hybrid TVSS

Hybrid suppression circuits as illustrated in Figure 21-7 will respond to several types of line disturbances. A gas tube is combined with a silicon clamping device to provide two-stage suppression.

When a transient hits, the gas tube fires and crowbars the bulk of the energy. The clamping device catches the leading edge of the transient that the gas tube may have missed.

The series impedance may be a resistor or an inductor sized to provide sufficient impedance to ensure the pulse will produce a high enough voltage to fire the gas tube.

21.8.3 Installation

The location of the surge suppressors has a large impact upon their efficiency. It is best if they are distributed, generally along the power distribution network. Figure 21-8 is an example of this staged protection.

21.9 Power

21.9.1 Source

Site planning should include an analysis of power source options such as utility purchased, third-party purchased, self-generated, and hybrid (a combination of sources). You should analyze each option in a way that considers cost, availability, capacity, and operating history.

You should also analyze the site to define its exposure to environmental disturbances (e.g., thunderstorms, tornadoes, earthquakes, hurricanes, and humidity). The EI should include the necessary design features to address these conditions, as needed, to achieve the appropriate availability. Options, for example, include steel versus wood poles, a counterpoise system for all overhead lines, underground lines instead of overhead lines, and parallel lines using a different physical routing.

The electrical distribution configuration (e.g., ring, dual parallel, or star) shall be suitable for the desired availability and be compatible with the design of the substations that service the process. Distribution substation configuration—their number, location, and capacity—should be compatible with the process distribution to minimize common-cause process failures, such as one substation loss

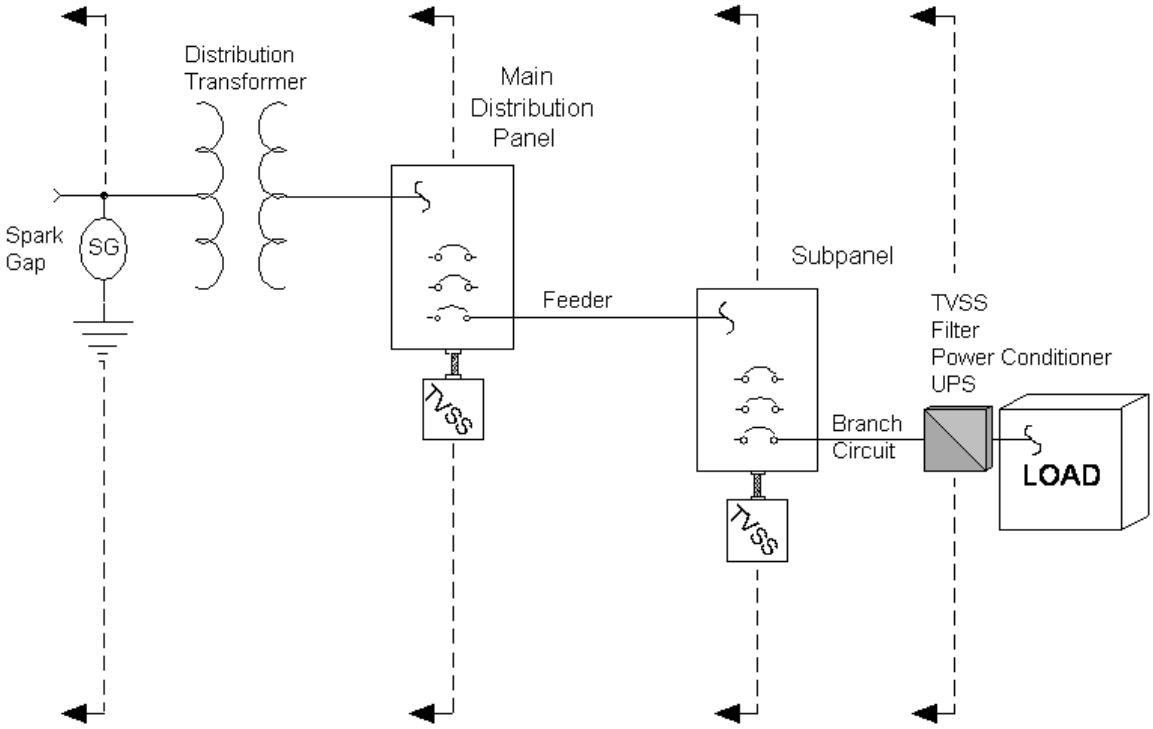


Figure 21-8: Staged TVSS

shutting down multiple process facilities. Electrical power voltage levels shall be compatible with the voltage required for the process electrical loads (e.g., three-phase motors/480 volts AC, single-phase motors/120 volt AC, PE device(s) power supply(s) input voltage/120 volt AC).

Electrical distribution metering is a key characteristic of any purchased power. For example, eight-cycle response time to transfer from one power source to another. What is the metering technology (e.g., carrier frequency on power lines, fiber optics, wireless)? What interface is required to the plant metering?

Is the metering provided compatible with the desired automation characteristics of the process? For example, will the process be able to withstand brownouts and voltage dips? Should process loads, such as pump motors, be provided with time-delay under-voltage protection to allow automatic start up after a voltage dip? What are the quantitative values used by the purchased power supplier for such things as voltage dips, voltage spikes, and brownouts? What impact will this have on the substation capacity? What impact will it have on the process?

A process may place heavy peak loads on the purchased power system. What effect does this have? For instance, the process may have a large motor (e.g., greater than 0.1500 HP) that starts up weekly. Energy needed to start the motor will lower the utility voltage. Will this level interfere with other customers? This analysis is required during preplanning to select the proper electrical power utility source.

Note that the contract agreed upon with the power supplier may stipulate maximum power demands the owner or user can place on the utility's power system. In any event, this value should be defined before EI design.

Electrical power systems for most PE automated processes require the ability to maintain power on portions of the control system for a fixed time during loss of utility power. Typical loads requiring this feature include:

- *PE logic solvers* such as programmable logic controllers (PLCs) and distributed control system controllers
- *sensor inputs* such as flow, pressure, temperature, and position switches
- *diagnostic outputs* such as indicating lights, alarms, and operator displays/keyboards

This power requires an alternative power source—for example, a fuel-powered electrical power generator or UPS—with the necessary speed of response and capacity to perform its function.

21.9.2 Power Quality

Power quality is a concept of powering equipment and devices in a manner suitable for the safe, reliable, and functional operation of that equipment. It requires monitoring and, if purchased, specifications that detail the amount of abnormality allowed.

21.9.2.1 Definitions of Abnormal Power

Sag – Also known as a dip, a sag is a reduction in RMS voltage at power frequency for a duration of a half-cycle to a few seconds.

Swell – An increase in RMS voltage at the power frequency from a half-cycle to 1 minute

Outage – A complete loss of power for a period of time, also known as a blackout

Impulse, Transient, or Surge – A subcycle disturbance; a sharp, brief discontinuity of the waveform of either polarity that may be additive or subtractive to the nominal waveform

Oscillatory Transient Or Ring Wave – A subcycle disturbance that rings or has the appearance of a decaying sine wave

Notch – A switching or other disturbance of the normal power voltage lasting less than a half-cycle. Initially opposite in polarity to the normal waveform creating a notch.

Noise – Unwanted electrical signal component either superimposed upon or modulated onto the desired signal

Harmonics – Frequencies that may exist on a power line that are multiples (e.g., 120, 180, 240, 300) of the fundamental frequency (60 Hz in the U.S.)

21.9.2.2 Harmonic characteristics

Nonlinear power devices cause harmonics. The measurement is total harmonic distortion (THD), and there are both voltage and current THDs. The ratio is the percent of the fundamental (line input) RMS value to the RMS value of the harmonic current or voltage.

Triplen harmonics are odd multiples of the third harmonic. These harmonics add when combined.

21.9.2.3 Problems Caused by Harmonics

- overheating of neutral
- overheating of transformers
- overheating of panel boards
- premature fuse or circuit breaker blowing
- poor power quality leading to data error and equipment malfunction
- resonant conditions in power system leading to large voltage transients

21.10 Uninterruptible Power Systems (UPS)

UPS come in many shapes forms and sizes depending upon:

- power required
- duration of power required
- switchover time
- power quality
- costs

UPS can be offline, online, or in another arrangement.

21.10.1 Offline UPS

Figure 21-9 is a block diagram of an offline UPS.

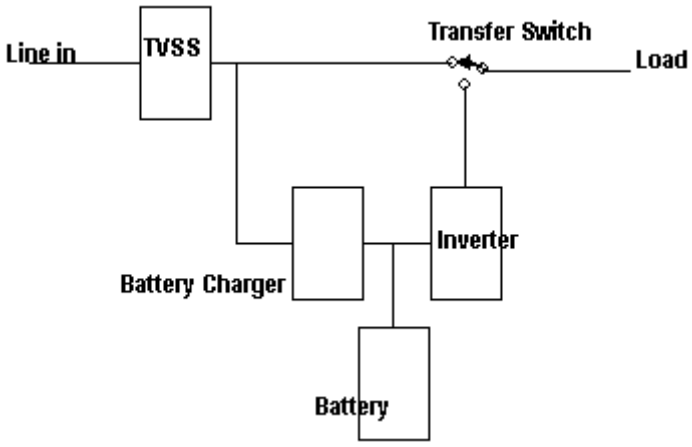


Figure 21-9: Offline UPS

The offline UPS is also called a standby or forward transferring UPS. It has a switching time of about 4 ms. This is generally not a problem with switch mode power supplies, because they have sufficient capacity to ride through the switching time. However, you will have to investigate the ability of linear power supplies to maintain voltage during the switching time; most industrial linear supplies should have little problem unless heavily loaded. A filter/TVSS must be present in the conduction path, because there is a direct path from line to load under normal operating conditions.

One consideration is the output wave shape. Many less expensive offline or standby UPSs output a square wave at about 90V (whose RMS value is 120V); it generates considerable noise, and you must consider the power quality factor. Generally, it is best to avoid a square wave output. On the way to a full sine wave, output is the stepped output allowing the load to integrate (smooth) into a quasi sine wave. More expensive units will produce a true sine wave output.

21.10.2 Online UPS

An online UPS is generally the supply for the process all the time. There are rotary models (using fly-wheels and motor/generators), and there are solid-state varieties. A block diagram of the solid-state type is illustrated in Figure 21-10. Note that this is an expensive solution, but one that ensures power quality, subcycle switchover, and (if solid state) no moving parts.

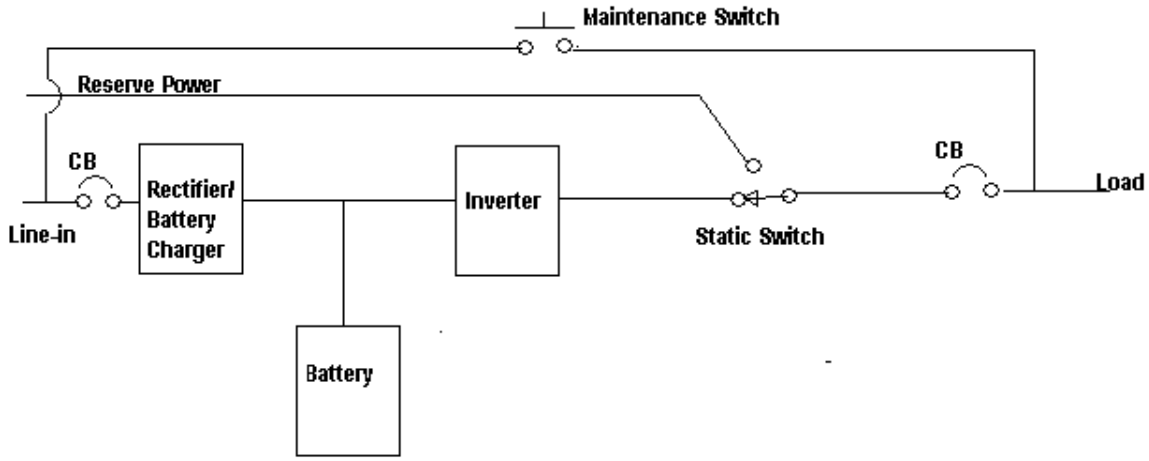


Figure 21-10: Online UPS

21.11 Electrical Installation Details

There are a myriad of installation details required for any EI. Pre-installation planning, process flow and control points, and documentation itself are absolute necessities. There are many differing requirements, and only some major electrical requirements are covered in the remainder of this chapter. The NEC and CEC dictate many of the wiring requirements that must be met for specific applications, including conductor capacity and sizing, conduit sizing, enclosure classifications, and protection characteristics and sizing.

21.11.1 Ampacity

NEC (2005) Section 310 *Conductors for General Wiring* defines and specifies the current carrying ability of conductors, their identification, and the derating necessary for specific conditions, while Annex C specifies how many conductors of which size can fit into conduits or raceways of a specified size.

21.11.2 Enclosures

Enclosures are required for various reasons, including protecting a person in the vicinity of electrical equipment that might prove hazardous.

Enclosures are required in hazardous areas to protect the area from the equipment. In hazardous areas equipment enclosures may be explosion proof. This means that even if there is an explosion in the cabinet, the cabinet is of sufficient strength and the portals to the areas outside the cabinet are of sufficient cooling length that ignition in the cabinet will not ignite the atmosphere outside the cabinet. Another cabinet use is that of a purge cabinet in which the cabinet is pressurized (usually with a non-incendive gas such as nitrogen), preventing ignition from taking place in the first place.

Enclosures protect equipment and personnel. See the National Electrical Manufacturers Association's (NEMA's) Web site (http://www.nema.com) for a list of standard enclosure specifications.

Enclosures are rated by their ability to protect. All provide protection against accidental exposure to the equipment inside (up to 1000V). However, they are then rated to withstand environmental conditions, such as indoors, outdoors, corrosion, or explosion proof. The IEC 60529 provides a set of specifications used in Europe and other countries, while in North America, NEMA has a set of standards. Please note that there is no direct correlation between the two specifications. However, in general, the more protections provided, the higher the cost of the enclosure.

One of the most popular types in the automation area is the NEMA Type 4. It is rated for indoors and outdoors; provides protection against dust, dirt, and water; and can even withstand external ice. Type 4X will provide all the above protections, as well as protect against corrosive agents. Type 6P provides all the protections of 4X and can also be submerged.

Types 7, 8, and 9 are for use in indoor hazardous locations, and type identifiers specify the class of materials. Type 10 is for methane gas (mines).

21.11.3 Protection

In this case, the term *protective devices* refers to devices such as circuit breakers, fuses, and ground fault current interrupters (GFCIs).

21.11.3.1 Fuses

Fuses are one-time overcurrent protection devices. The active element melts when excessive current flows through the element. There are single element types (quick blow) and dual element types (slo-blo).

Fuses come in many forms and have four ratings: (1) voltage, (2) amperage, (3) current interrupting—AC Short Circuit [AIR]—rating, and (4) instantaneous rating. Please note that cartridge fuses above a certain amperage can no longer have the round end but must use blades for contacts.

21.11.3.2 Fuse Rejection Feature

Table 21-3: Rejection Lengths of Fuses

Amperage	250V Length	600V length
0.1 – 30	2 in	5 in
35 – 60	3 in	5.5
70 – 100	5 7/8 in	7 7/8
110 – 200	7 1/8 in	9 5/8
225 – 400	8 5/8 in	11 5/8
450 – 600	10 3/8 in	13 3/8

The fuse rejection feature is used on fuses so lower AIR rated fuses cannot be physically substituted for higher rated ones.

2.11.3.3 Circuit Breakers

Circuit breakers are typically one of two types: bimetallic or magnetic.

Bimetallic types use a bimetallic strip (two metals of different temperature coefficients of expansion bonded together) to release or otherwise cause the breaker to trip. The bimetallic types have fair repeatability (relative to the other types), but ambient temperature must be considered because it causes lower trip points at higher temperatures.

Magnetic types are found in industry and use the line current to pull in and unseal a line experiencing overcurrent. They have trips that (in some cases) may be adjusted.

2.11.3.4 GFCI

GFCI devices compare the current in the phase line and the neutral. Any imbalance means the current has found a different path to return, and at the trip rating of the GFCI will open the circuit, preventing possible electric shock.

These should be located in an industrial control system anywhere water is collocated with the power. Typically used where detection of minute (compared to a fault) current paths exist to ground to prevent electrical shock.

The typical residence type triggers on an imbalance of 4 mA, the industrial types at about 10 mA or, if adjustable, within their range.

21.12 References

ISA

ISA-5.1-1984 - (R1992) – *Instrumentation Symbols and Identification.*

ISA-12 Series of Standards, Recommended Practices (RP), and Technical Reports (TR) related to Electrical Equipment for Hazardous Locations.

ISA-50 Series of Standards and Technical Reports related to Signal Compatibility of Electrical Instruments.

ISA-60 Series of Recommended Practices (RP) for Control Centers.

ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1-3 Mod) – *Functional Safety: Safety Instrumented Systems for the Process Industry Sector.*

IEEE

BSR/IEEE 62 Series of Guides and Standards for Surge Protection.

IEEE 142-1991– *Recommended Practice for Grounding of Industrial and Commercial Power Systems*. (IEEE Green Book.)

IEEE 519-1992 – *Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems*.

IEEE 1100-1999 – *Powering and Grounding Sensitive Electronic Equipment*.

IEEE 1159-1995 – *Recommended Practice for Monitoring Electric Power Quality.*

IEEE Standards Reference – An enhanced InfoBase version of the *IEEE Standard Dictionary of Electrical and Electronics Terms* on CD-ROM (formerly IEEE 100).

National Fire Protection Association

NFPA 70 – *National Electric Code® - 2005 edition.*

NFPA 79 – *Electrical Standard for Industrial Machinery - 2002 edition.*

Other

MIL-HDBK 419A – *Grounding, Bonding, and Shielding for Electronic Equipments and Facilities.*

About the Authors

Victor J. Maggioli Sr., a 38-year DuPont veteran and currently President, Feltronics Corp., Newark, Del., has made significant contributions to the process industry sector by applying new techniques in the areas of grounding, power line conditioning, programmable electronics, fiber optics, fail-to-safe configured control systems, and limited variability application software. A member of the European Workshop for Industrial Computers (EWICS), he co-authored a guideline on *Use of PLCs in Process Safety Applications*, which was issued by EWICS. Maggioli chaired ISA standards panel 84 (SP84) and led a committee of more than 100 engineers in the development of ANSI/ISA-84.00.01-1996, and ISA-TR84.00.02. He also participated in the International Electrotechnical Commission (IEC) develop-

ment of the generic functional safety standard IEC 61508 and was selected to lead the process sector standard development (i.e., IEC 61511) for functional safety using PE technology, which resulted in approval of IEC 61511. He has been a member of ISA, IEEE, AIChE, and NFPA. Maggioli is a senior member in ISA and a life senior member in IEEE. Maggioli received the 1996 DuPont Engineering Excellence Award for outstanding contribution in the area of process control functional safety and *CONTROL* magazine's 1987 engineer of the year award for his contribution in process safety standards development. He serves as an ANSI-appointed technical representative on industrial computer safety to the IEC.

Graham Elvis earned a post-graduate diploma in Machine Tool Design and Technology from the University of Manchester (UK) Institute of Science and Technology. He has worked for Rockwell Automation for 27 years in various posts involving Drive System Technology and Industrial Automation. Currently, he is a senior Customer Success Team member.

Lawrence (Larry) M. Thompson has an extensive background as a technician, technical trainer, and course developer in electronics, measurement and control, and computer networking. A 20-year veteran of the U.S. Air Force, his military specialty included instruction in electronic encryption equipment and TEMPEST. He has maintained a General (formerly First Class) FCC Radiotelephone license since 1966. His industrial experience includes positions as technician, test engineer, and test engineering supervisor for numerous companies. He recently retired as Department Chair for Software Engineering Technology at Texas State Technical College, and now runs his own consulting business. He has served as an adjunct instructor for ISA for more than 20 years. He has a B.A.A.S. from Tarleton State University, Texas, and is the author of several books including ISA's *Industrial Data Communications* and *Basic Electricity and Electronics for Control: Fundamentals and Applications.*